Evaluation and Comparison of Graph Summarization Methods

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Abstract

As real-world networks grow to billions of nodes and edges, performing even basic analyses – such as diffusion, community detection, and shortest-path queries – becomes computationally prohibitive. Graph summarization offers a principled solution by constructing a much smaller surrogate graph G' that faithfully preserves the analytical properties of an original graph G. In this paper, we focus on three broad classes of summarization techniques – spectral sparsification, community collapse, and spectral coarsening – each tailored to preserve a distinct class of graph properties (spectral, meso-scale, and diffusion, respectively). We first explore the mathematical intuition and theoretical performance bounds for each paradigm, then introduce a unified evaluation framework comprising spectral approximation error, community-structure fidelity, distance distortion, centrality retention, and compression ratio. Through extensive experiments on four large web graphs, we demonstrate that (1) community collapse consistently achieves the lowest spectral error and smallest average-path distortion but can degrade community fidelity when over-compressed, (2) spectral sparsification is the fastest to summarize, delivers the highest community-structure retention, and provides balanced spectral and distance preservation, and (3) spectral coarsening yields the best PageRank top-k retention and moderate performance across all metrics at moderate compression, at the cost of increased runtime. Our results validate the theoretical trade-offs and offer task-driven guidelines for selecting – or combining – summarization strategies based on the desired fidelity and computational constraints.

1 Introduction

In the era of big data, real-world graphs – ranging from social networks and communication graphs to web-scale hyperlink structures – often consist of hundreds of millions to billions of nodes and edges. Running even basic graph algorithms (e.g., PageRank, community detection, shortest-path computations, etc...) on such massive structures can be prohibitively expensive in both time and memory. Graph summarization addresses this challenge by constructing a much smaller surrogate graph $G' = (V', E')$ from an original graph $G = (V, E)$, such that $|V'| \ll |V|$, $|E'| \ll |E|$, yet G' preserves the essential analytical properties of G.

What makes a good summary graph? A summary graph must faithfully replicate the behaviors of the original in ways that matter for downstream applications. Since different analyses depend on different structural features, we focus on three broad classes of properties:

1. *Spectral properties:* These include quadratic forms of the graph Laplacian L_G, which underpin diffusion processes, random-walk mixing times, and many centrality measures (e.g., eigenvector centrality). Approximating spectral behavior ensures that algorithms

Code repository available at this link: https://github.com/evanjiang943/cs2241.

relying on eigenvalues or Laplacian-based embeddings (such as PageRank or spectral clustering) yield similar results on G' as on G.

2. *Community (meso-scale) properties:* Many graphs exhibit modular structure, where nodes form tightly interconnected groups or communities. Preserving community structure is crucial for tasks such as role discovery, recommendation systems, or any analysis that leverages the presence of densely connected subgraphs.
3. *Distance (diffusion) properties:* Applications like routing, influence maximization, or network robustness depend on shortest-path distances or random-walk hitting times. Ensuring that path lengths and diffusion dynamics are similar in G' and G guarantees that distance-based analyses (e.g., betweenness centrality or epidemic simulations) remain reliable.

Focusing on these three classes is motivated by the fact that most graph-analytics pipelines revolve around one or more of these dimensions: spectral methods for global structure, community detection for modular insights, and distance measures for connectivity and flow. By designing summaries that target each class, we can cover the vast majority of practical use cases, and also compare how different summarization paradigms trade off among these desiderata.

These desiderata naturally give rise to three main paradigms for graph summarization:

1. Spectral Sparsification (Sampling) Spectral sparsifiers approximate L_G by a weighted subgraph $G' = (V, E')$ of much smaller size, ensuring for all $\mathbf{x} \in \mathbb{R}^{|V|}$:

$$(1-\varepsilon)\,\mathbf{x}^\top L_G\,\mathbf{x} \;\le\; \mathbf{x}^\top L_{G'}\,\mathbf{x} \;\le\; (1+\varepsilon)\,\mathbf{x}^\top L_G\,\mathbf{x}.$$

Sampling edges based on effective resistance R_e and re-weighting them by $1/p_e$ preserves the global spectral signature of G [1]. This ensures that diffusion-based measures and Laplacian eigenvalues remain stable.

2. Community-Collapse (Clustering) Clustering-based summarization algorithms identify communities $C_1, \ldots, C_k \subseteq V$ (e.g., via Louvain, Infomap) by optimizing a quality function such as modularity:

$$Q = \frac{1}{2m} \sum_{i,j} \Big[A_{ij} - \frac{d_i d_j}{2m}\Big] \delta(c_i, c_j).$$

Collapsing each detected community into a supernode preserves the meso-scale organization of G, ensuring that intra- and inter-community connectivity patterns are reflected in G' [2].

3. Spectral Coarsening (Embedding) Spectral-coarsening computes the first k eigenvectors of the normalized Laplacian L_{norm}, embedding nodes into a k-dimensional space. Grouping nodes in this spectral domain (e.g., via k-means) and collapsing them maintains diffusion distances and random-walk dynamics, since the leading eigenvectors capture the principal modes of variation in G [3].

Contributions. Although spectral sparsification, community collapse, and spectral coarsening each offer provable guarantees for a particular class of graph properties, practitioners lack clear guidance on how they compare across multiple fidelity dimensions and runtime budgets. In this work, we:

- Propose a unified evaluation framework that motivates and measures five core fidelity metrics – spectral approximation error, community-structure retention, distance distortion, centrality retention, and compression ratio – alongside summarization and evaluation runtime.
- Explore mathematical intuitions and theoretical performance bounds for each paradigm, setting expectations for how edge sampling, node aggregation, and spectral embedding should behave.
- Implement state-of-the-art versions of each method and conduct an extensive empirical study on four large web graphs, showing that community collapse minimizes spectral error and distance distortion, spectral sparsification is fastest and best preserves community labels, and spectral coarsening balances all metrics, while examining expected results in the data.

Sections follow this structure: Section 2 reviews related algorithms and metrics and also details our evaluation protocol and theoretical expectations; Section 4 presents the experimental results; Section 5 synthesizes data and insights; Section 6 discusses limitations in our approach, and Section 7 outlines avenues for future work.

2 Related Work

Building on the desiderata outlined in the introduction, prior work on graph summarization naturally falls into three paradigms – spectral methods, community-based compression, and spectral embedding/coarsening – and a growing body of evaluation frameworks that measure summary fidelity across multiple metrics.

Spectral sparsification and coarsening methods seek to preserve the global spectral and cut structure of a graph by judiciously sampling or merging edges and nodes. Spielman and Srivastava pioneered effective-resistance sampling to obtain an $O(n \log n/\varepsilon^2)$-edge sparsifier whose Laplacian quadratic forms satisfy $(1 \pm \varepsilon)$ bounds for all vectors x [1]. Loukas extended this line of work by unifying sparsification and node-merging under a restricted spectral approximation framework, yielding algorithms that coarsen or sparsify while provably preserving both eigenvalues and cut capacities [3]. Empirical studies confirm that these spectral techniques maintain connectivity, eigenstructure, and diffusion dynamics, making them the methods of choice when downstream tasks rely on random-walk behavior or centrality measures. However, although both works provide illustrative examples, their empirical scope is intentionally narrow. For example, Spielman and Srivastava focus solely on theoretical bounds, while Loukas' experiments are limited to small graphs with less than 3,000 nodes and 70,000 edges, without exploring web-scale data, diffusion/centrality tasks, or multiple compression settings beyond the few ratios reported.

Community-based summarization methods instead group nodes into "supernodes" to capture meso-scale modular structure. Navlakha *et al.* introduced a two-part summary – an aggregate supernode graph accompanied by edge-correction lists – optimized via Minimum Description Length to achieve tunable lossy compression with explicit reconstruction-error bounds [2]. However, because the MDL formulation requires a combinatorial search over partitions and can generate correction lists whose total size approaches that of the original graph, it becomes prohibitively expensive both in time and space once a graph exceeds a hundred-thousand nodes. Later, Riondato *et al.* recast this problem as a k-means clustering on the adjacency matrix, providing a constant-factor approximation algorithm that minimizes L_1 or L_2 reconstruction error and significantly outperforms earlier heuristics such as GraSS [4]. These approaches excel at preserving community modularity and high-level connectivity patterns under aggressive size reduction. Unfortunately, the authors ultimately concluded that "for many graphs, k-summaries may achieve such a low reconstruction error only for very high values of k, often close to n," suggesting that results can end up with as many supernodes as original nodes, defeating the purpose of summarization.

A handful of studies have conducted cross-paradigm empirical comparisons, highlighting inherent trade-offs and motivating unified evaluation. Chen *et al.* benchmarked twelve sparsifiers – including effective-resistance, random pruning, and degree-based heuristics – across sixteen metrics spanning spectral error, cut distortion, centrality and community fidelity, clustering coefficient, and downstream machine-learning performance, demonstrating that no single method dominates all criteria [5]. Liu *et al.* compared summaries generated by different community detection algorithms using MDL-based metrics (edge overlap, pattern coverage), showing that highest modularity does not necessarily yield the most concise or informative summary [6]. In fact, they went on to find that no single clustering approach simultaneously maximizes compression, coverage, and efficiency, thereby motivating the need for a unified summarization framework.

To support such comparisons, a suite of evaluation metrics has emerged. Spectral and cut similarity metrics measure eigenvalue deviations, quadratic-form bounds, and worst-case cut-weight distortion [1], [3]. Distance preservation is assessed via shortest-path stretch or All-Pairs Shortest Path distortion [7]. Centrality retention uses rank correlation or top-k overlap for PageRank and related measures, as in FrogWild!'s PageRank approximation guarantees [8]. Community structure fidelity is quantified by modularity retention, normalized mutual information between partitions, and conductance. Finally, task-based metrics evaluate query accuracy, speed-up, or end-to-end performance on graph-learning tasks [5]. Survey work by Liu *et al.* advocates reporting summary size alongside these multi-dimensional distortion measures to enable fair cross-method comparisons [9].

Our work extends these foundations by unifying spectral, community, and distance metrics into a single evaluation framework, deriving theoretical performance bounds for each summarization paradigm, and empirically guiding the selection of the most suitable summarization strategy for diverse graph-analysis tasks.

3 Metric Selection and Theoretical Expectations

Evaluating graph summaries requires balancing the fidelity of structural properties with the computational savings achieved through node and edge reduction. To capture the breadth of real-world graph-analysis tasks, we select five complementary metrics that together cover the most critical dimensions of summary quality: global spectral fidelity, meso-scale community structure, local connectivity, node-level importance, and raw size reduction. By considering this suite of metrics in concert, we can characterize both the strengths and limitations of different summarization paradigms.

Overview of Metrics

1. Spectral Approximation Error Global analyses on graphs, such as PageRank, spectral clustering, and diffusion processes, fundamentally rely on the spectrum of the graph Laplacian. Distortions in eigenvalues disrupt mixing times and eigenvector-driven methods. We measure

$$\mathrm{SpectralErr} = \frac{1}{k} \sum_{i=2}^{k+1} \left| \frac{\lambda_i' - \lambda_i}{\lambda_i} \right|, \tag{1}$$

where λ_i are the nontrivial eigenvalues of the normalized Laplacian of the original graph G, and λ_i' the corresponding values for the summary G'.

Effective-resistance sparsification provides strong $O(\varepsilon)$ multiplicative guarantees on all eigenvalues up to the chosen rank, ensuring that $\mathrm{SpectralErr} = O(\varepsilon)$ even for small summaries. Spectral coarsening bounds this error in terms of the spectral mass discarded beyond the top k eigenmodes, i.e. $O\big(\frac{1}{k}\sum_{j>k}\lambda_j\big)$, and thus performs well when the original graph has rapidly decaying eigenvalues. In contrast, community collapse lacks explicit spectral guarantees and can yield larger errors, particularly when clusters do not align with the dominant eigenspace.

2. Community-Structure Fidelity Many graph tasks – such as recommendation systems, role discovery, and anomaly detection – depend on identifying tightly-connected groups (communities). We compare the ground-truth community labels C on G with those C' on G' using normalized mutual information:

$$\mathrm{NMI}(C, C') = \frac{2, I(C; C')}{H(C) + H(C')}, \tag{2}$$

where $I(\cdot;\cdot)$ is the mutual information and $H(\cdot)$ the entropy.

Community collapse trivially achieves $\mathrm{NMI} = 1$ by construction, as it treats each base community as an atomic summary node. Spectral coarsening can approach perfect fidelity when natural community divisions coincide with large eigengaps, but may degrade if small communities are absorbed into larger spectral clusters. Effective-resistance sparsifiers, while preserving global connectivity patterns, may remove intra-community edges and thus produce moderate NMI values depending on the choice of ε.

3. Distance Distortion Finer-grained connectivity is critical for applications such as routing, reachability queries, and influence spread. We sample a set S of node pairs and report the average stretch:

$$\mathrm{Stretch} = \frac{1}{|S|} \sum_{(u,v)\in S} \frac{d'uv}{duv}, \tag{3}$$

where d_{uv} and d'_{uv} denote shortest-path distances in G and G', respectively.

By design, effective-resistance sparsifiers guarantee low distortion for distances tied to edge resistances, often incurring at most $O(\log n)$ stretch. Spectral coarsening tends to preserve small-scale distances within each spectral cluster, yielding stretch near $1 + O(\sum_{j>k} \lambda_j / \lambda_2)$, but may distort longer paths across clusters. Community collapse typically underestimates intra-community distances (stretch < 1) due to node aggregation, while potentially inflating inter-cluster paths significantly.

4. Centrality Retention Identifying influential nodes (hubs) is pivotal in marketing campaigns, epidemic control, and network robustness analysis. Let π and π' be the PageRank vectors on G and G', respectively. We define

$$\text{Precision@}k = \frac{|\text{top-k}(\pi) \cap \text{top-k}(\pi')|}{k}. \tag{4}$$

Sparsifiers with ε-resistance guarantees perturb PageRank by at most $O(\varepsilon/(1-\alpha))$, preserving the identity of most top-k nodes for sufficiently small ε. Spectral coarsening maintains the dominant eigenspace corresponding to the personalized teleportation submatrix, yielding high precision when k is chosen below the coarsening rank. Community collapse can misrank bridge nodes whose importance arises from inter-community links, reducing Precision@k unless k is large relative to the number of communities.

5. Compression Ratio The ultimate goal of summarization is to reduce computation and memory usage. We quantify size reduction as

$$\text{CR} = \frac{|V'| + |E'|}{|V| + |E|}, \tag{5}$$

reflecting the combined node and edge counts of the summary relative to the original. Most graph algorithms scale roughly linearly in $|V| + |E|$, so a summary with $\text{CR} = 0.1$ typically consumes $\approx 10\%$ of the memory and runtime for linear-time routines. For superlinear algorithms – e.g. eigenvalue solvers or community-detection heuristics – reducing problem size can yield superlinear speed-ups. Thus, including CR alongside fidelity metrics illuminates the trade-off: lower CR implies greater efficiency gains at the potential cost of higher spectral, community, distance, or centrality error.

Theoretical Expectations Across Summarizers

Given these metrics, we outline our hypotheses for three summarization paradigms:

- **Spectral Sparsification (SS):** We expect SS to excel at preserving spectral and centrality metrics for small ε, while achieving moderate community fidelity and distance distortion. Compression ratios can be tuned via ε, with CR decreasing as ε grows.

- **Spectral Coarsening (SC):** SC should offer strong performance on spectral error, community fidelity (when communities align with eigenmodes), and moderate distance retention within clusters. CR is determined by the chosen coarsening rank k, balancing fidelity versus reduction.

- **Community Collapse (CC):** CC guarantees perfect community-structure fidelity and offers strong CR when communities are small. However, CC often introduces large spectral and distance errors, and may misrank bridging nodes in centrality measures.

By evaluating these expectations in Section 4.3, we aim to provide practitioners with concrete guidance on selecting summarization techniques tailored to their analytic priorities.

4 Results

4.1 Web Graph Benchmarks

To evaluate the real-world applicability of our summarization framework, we focus on four canonical large-scale web graphs: `web-Google`, `web-Stanford`, `web-BerkStan`, and `web-NotreDame`. These datasets, which have been extensively studied in both academic and applied settings, provide complementary structural challenges for summarization techniques.

Scale and Sparsity. The `web-Google` graph comprises approximately 875,000 nodes and 5.1 million directed edges, capturing an extensive snapshot of real-world hyperlink topology. In contrast, `web-Stanford` (282,000 nodes, 2.3 M edges), `web-BerkStan` (685,000 nodes, 7.6 M edges), and `web-NotreDame` (326,000 nodes, 1.5 M edges) span a spectrum of sizes and densities. Despite differing scales, all four graphs maintain a comparably low average degree, reflecting web sparsity and underscoring the need for efficient edge- and node-level reduction strategies.

Global Statistics. Beyond simple node and edge counts, all four graphs exhibit small-world properties (diameter $\ll |V|$) and heavy-tailed shortest-path length distributions. For example, `web-Google` has an average path length of approximately 6 hops despite its 875 K nodes, reflecting the dense "backbone" formed by high-throughput hubs. By contrast, `web-NotreDame`—with only 1.5 M edges—has a larger diameter and a more skewed path-length distribution, highlighting its more hierarchical site layout.

Local Connectivity Patterns. Each web graph has a characteristic local clustering-coefficient distribution. `web-Stanford` shows a pronounced bimodal pattern, with research-group pages forming high-clustering "cliques" and administrative pages forming sparse link-chains. In contrast, `web-BerkStan` contains many "hub-and-spoke" motifs—central index pages linking to dozens of topic pages—demanding that summaries preserve both hubs (for reachability) and spokes (for topical browsing).

Degree Distributions and Rich Connectivity. Each graph exhibits a power-law degree distribution, whereby a small fraction of hubs command a disproportionately high number of incoming and outgoing links. Effective summarization must preserve both global connectivity patterns and the nuanced roles of hubs in facilitating cross-community flow.

Community Structure and Modularity. Pronounced modularity is present in each dataset, with tightly knit clusters corresponding to thematic subgraphs (e.g., departmental pages in Stanford, research groups at Notre Dame). Summaries should maintain high intra-community cohesion while accurately approximating inter-community connections.

Directed Navigation Patterns. The directed nature of hyperlinks captures realistic user or crawler pathways. Preserving edge directionality is critical for downstream applications such as PageRank approximation, random-walk simulations, and navigation modeling.

Alignment with Prior Literature. All four datasets are publicly available via the SNAP repository. By selecting these widely used benchmarks, we align our evaluation with a large body of existing work on graph sampling, sparsification, and coarsening. This choice ensures our fidelity metrics – spectral error, community NMI, average stretch, precision@k, and compression ratio – are directly comparable to established results.

Experimental Protocol. For each graph, we apply three summarization techniques:

- **Sparsifier**: Spectral sparsification to remove edges while preserving key spectral properties.
- **Collapse**: Community collapse to merge nodes within detected clusters.
- **Coarsener**: Spectral coarsening to group nodes based on spectral embeddings.

We vary reduction parameters systematically to explore trade-offs between summary compactness and fidelity, reporting five core fidelity metrics alongside summarization and evaluation runtimes. Through this design, we highlight how each method handles sparsity, hub dominance, modularity, and directionality in large-scale web graphs.

4.2 Experimental Setup

Table 1 summarizes the four directed web-graph datasets used in our experiments.

Table 1: Dataset statistics.

Graph	#Nodes	#Edges	Description
web-Google	875,713	5,105,039	Google web crawl
web-Stanford	281,903	2,312,497	Stanford.edu web graph
web-BerkStan	685,230	7,600,595	Berkeley & Stanford crawl
web-NotreDame	325,729	1,497,134	Notre Dame website

We implemented all three summarization paradigms in Python, leveraging NetworkX for graph data structures and common scientific libraries for numerical operations. Each summarizer extends a common `GraphSummarizer` base class that handles timing, I/O, and summary-graph bookkeeping; the core logic for community, spectral coarsening, and spectral sparsification is encapsulated in three modules as described below.

4.2.1 Community-Based Collapse

Our community-collapse implementation (`collapse.py`) uses the Louvain method (`python-community` package) to group nodes into supernodes:

- **Directed handling:** Input `DiGraph`s are symmetrized (`to_undirected()`) so that Louvain can be applied.
- **Partitioning:** Call `community_louvain.best_partition(graph, resolution, random_state=42)` to obtain a mapping $v \mapsto c(v)$.
- **Summary graph construction:**
  1. Create one summary node per community and add `size`, `members`, and `internal_edges` attributes.
  2. For each original edge $\{u, v\}$, if $c(u) \neq c(v)$, aggregate inter-community weights and counts.
  3. Normalize each summary edge by the maximum possible connections $|C_u| \times |C_v|$ to record a `density` attribute.
- **Complexity:** Louvain runs in $O(m \log n)$ on sparse graphs; summary-graph construction is $O(m)$.

4.2.2 Spectral Coarsening

Our spectral-coarsener (`spectral_coarsener.py`) clusters nodes in a low-dimensional Laplacian eigenspace:

- **Laplacian computation:** We form either the normalized or unnormalized Laplacian via NetworkX's sparse routines.
- **Eigenpair extraction:**
  - For $n > 5000$, we use SciPy's `lobpcg` on the CSR Laplacian, initializing with random vectors and a max-iteration budget.
  - For smaller graphs, we fall back to a dense `np.linalg.eigh` call.

- **Spectral embedding:** Sort eigenvalues ascending, skip the trivial constant eigenvector, and form a feature matrix from the next k eigenvectors, where $k = \min(\text{target size}, n-1)$.
- **Clustering:** Apply `MiniBatchKMeans` with batch-size $\sim$1–5% of n, a tight convergence tolerance, and a single initialization for speed.
- **Summary graph:**
  1. Map each node to its cluster ID, create supernodes with a `size` attribute.
  2. Aggregate inter-cluster edges (summing weights and counts).
  3. Record `internal_edges`, `max_connections`, and `density` for each summary edge.
- **Complexity:** Dominated by $O(m\,k\,T)$ for eigenvector computation and $O(n\,k\,T')$ for clustering, where T and T' are iteration counts.

4.2.3 Spectral Sparsification

Our sparsifier (`spectral_sparsifier.py`) implements effective-resistance sampling to sparsify edges:

- **Node retention:** All original nodes are kept; the summary graph has the same node set.
- **Resistance estimation:**
  - For $n \leq n_{\max}$ (default 1000), compute the Laplacian pseudoinverse to get exact resistances.
  - Otherwise, approximate $R_{uv} \approx (1/d(u) + 1/d(v))/w_{uv}$ using degree heuristics.
- **Sampling probabilities:** Set $p_{uv} \propto w_{uv}\,R_{uv}$ and normalize.
- **Edge sampling:** Draw $m' = \max(n-1, \lfloor m\alpha \rfloor)$ edges with replacement, then reweight each sampled edge by $\frac{w_{uv} \cdot (\text{count})}{p_{uv}\,m'}$.
- **Summary graph:** Insert sampled edges (with attributes `weight`, `original_weight`, `sampled_count`) into the summary.
- **Complexity:** $O(m)$ to compute approximate resistances, $O(m' \log m)$ for sampling, and $O(m')$ for reconstruction.

All three summarizers share a common evaluation interface, allowing them to be easily swapped when computing our five core fidelity metrics. This modular design ensures that implementation details (e.g. choice of eigensolver or clustering algorithm) can be tuned independently while preserving a consistent experimental protocol.

4.3 Experimental Results

We evaluate each summarization method on four web graphs. Tables 2, 3, and 4 report the five core fidelity metrics plus summarization and evaluation times.

Dataset	SpErr	CommNMI	AvgStr	P@50	CompR	SummT (s)	EvalT (s)
web-BerkStan	0.471	0.427	0.150	0.54	0.00039	1,279.3	87.7
web-Google	0.454	0.334	0.104	0.54	0.00206	317.2	108.5
web-NotreDame	3.204	0.154	0.174	0.40	0.00157	134.5	28.0
web-Stanford	0.960	0.491	0.157	0.84	0.00078	739.2	31.1

Table 2: Performance of the *collapse* summarizer.

5 Discussion

Our experiments on four large web graphs confirm the theoretical trade-offs among the three summarization paradigms and demonstrate how practitioners can choose a method based on their specific fidelity requirements.

Dataset	SpErr	CommNMI	AvgStr	P@50	CompR	SummT (s)	EvalT (s)
web-BerkStan	0.965	0.205	0.258	1.00	0.27423	5,115.1	185.5
web-Google	0.951	0.131	0.249	0.78	0.46182	1,635.1	213.2
web-NotreDame	0.940	0.430	0.429	0.70	0.27748	241.3	28.3
web-Stanford	0.864	0.209	0.281	0.80	0.34539	109.3	36.6

Table 3: Performance of the *coarsener* summarizer.

Dataset	SpErr	CommNMI	P@50	CompR	SummT (s)	EvalT (s)
web-BerkStan	1.000	0.733	0.66	0.16278	26.0	108.5
web-Google	0.778	0.864	0.82	0.27006	27.1	141.9
web-NotreDame	1.459	0.844	0.62	0.31627	6.1	31.3
web-Stanford	1.429	0.750	0.58	0.20360	8.8	33.9

Table 4: Performance of the *sparsifier* summarizer (Stretch undefined).

Compression Regimes. As intended, the three methods occupy distinct regions of the compression–fidelity spectrum. The community-collapse summarizer achieves extreme compaction (CompR on the order of 10^{-4}), drastically reducing both nodes and edges by merging densely connected modules. The spectral sparsifier provides moderate edge reduction (CompR ≈ 0.16–0.32) while preserving all original nodes, and the spectral coarsener retains a significant fraction of nodes and edges (CompR ≈ 0.27–0.46) through clustering in a low-dimensional embedding. These size differences give users immediate guidance on which paradigm to apply when storage or processing budgets vary by orders of magnitude.

Spectral Fidelity. We expected community collapse to minimize spectral distortion by preserving quadratic forms within each module, and indeed it yields the lowest spectral error (SpErr ≤ 0.96) on three of four datasets (Table 2). The exception is web-NotreDame (SpErr = 3.20), where collapsing only 570 nodes hurts fine-grained spectral detail – highlighting a practical lower limit on compaction. As predicted, the coarsener exhibits the highest spectral error (SpErr ≈ 0.86–0.98, Table 3), consistent with the truncation bound $O(\sum_{j>k} \lambda_j)$. The sparsifier falls in between (SpErr ≈ 0.78–1.46, Table 4), its effective-resistance sampling guarantee holding except on the smallest graph where edge removal has disproportionate effect. This hierarchy – collapse $<$ sparsifier $<$ coarsener – aligns closely with our theoretical analysis in Section 3.

Community Preservation. Community-structure fidelity follows the inverse ordering. The sparsifier most faithfully retains Louvain partitions (CommNMI ≈ 0.73–0.86), since edges within communities tend to have low effective resistance and remain in the summary. Collapse obtains moderate NMI (0.15–0.49), reflecting that supernodes may encompass multiple original communities. Coarsening performs worst (NMI ≈ 0.13–0.43), as spectral clusters do not necessarily coincide with modularity-driven groups (Table 3). These results reinforce that preserving meso-scale structure requires targeted edge or node aggregation, not pure spectral reduction.

Distance Distortion. Consistent with our expectations, collapse produces the smallest average-stretch (AvgStr ≈ 0.10–0.17, Table 2) by collapsing all intra-community distances to one, thus underestimating true path lengths but maintaining relative locality. The coarsener shows higher AvgStr (0.25–0.43, Table 3), increasing with the fraction of retained nodes and matching the theoretical bound $1 + O(\sum_{j>k} \lambda_j / \lambda_2)$. We did not measure distortion for the sparsifier, but prior work suggests potential logarithmic stretch in the worst case.

Average Stretch and Sparsifier Limitations. Average stretch quantifies how much shortest-path distances change in the summary. For community collapse, intra-community pairs collapse to distance 1, yielding very low stretch (0.10–0.17), while spectral coarsening obeys the theoretical bound $1 + O\left(\frac{\sum_{j>k} \lambda_j}{\lambda_2}\right)$, and we observe stretch in the range 0.25–0.43 as expected.

We do not report stretch for the spectral sparsifier, because removing edges can disconnect node pairs, which renders the ratio d'_{uv}/d_{uv} undefined under the usual definition. Moreover, effective-resistance sampling guarantees only spectral-norm preservation of the Laplacian quadratic form – not bounded path-length distortion – so a meaningful stretch estimate would require special handling of disconnected pairs (e.g. excluding them or capping their contribution) and combining sparsification with spanner techniques to obtain formal worst-case stretch bounds. In our context, this complexity makes sparsifier-based stretch measurements neither straightforward nor directly comparable to the other summarizers.

Centrality Retention. Clustering in the leading-eigenvector space gives the coarsener superior PageRank precision (P@50 up to 1.00, Table 3), confirming that the summary preserves the dominant modes of diffusion. The sparsifier follows (P@50 ≈ 0.66–0.82, Table 4), in line with perturbation bounds $\|\pi' - \pi\|_1 = O(\varepsilon/(1-\alpha))$. Collapse is the weakest (P@50 ≈ 0.40–0.84, Table 2), since merging hubs with their neighbors dilutes their relative importance. These results direct users to select coarsening for hub-focused analyses, sparsification for a middle ground, and collapse when centrality is less critical.

Runtime and Scalability. Empirical runtimes highlight practical constraints: sparsification is fastest to execute (25–30s summarization, 100–142s evaluation), making it suitable for interactive or large-scale applications. Collapse requires moderate time (134–1,279s summarization, 28–108s evaluation), dominated by Louvain community detection. Coarsening is the most expensive (241–5,115s summarization, 29–213s evaluation), due to eigenvector extraction and minibatch clustering. This confirms that only the sparsifier scales easily to streaming or real-time contexts, while collapse and coarsener are better suited to offline batch processing.

Synthesis and Recommendations. No single summarization method dominates all metrics; rather, each excels along one or two axes:

- Use *collapse* for maximum spectral and distance fidelity under extreme compaction – avoiding over-compression on small graphs.
- Use *sparsifier* for rapid summaries with balanced preservation of spectral, community, and centrality properties.
- Use *coarsener* when top-k centrality accuracy is paramount and computational resources permit eigenvalue computations.

By mapping each method onto this performance landscape, we realize the paper's vision of a unified, theory-informed framework that guides summarization strategy selection based on concrete fidelity and runtime metrics.

5.1 Unexpected Observations

While most results matched our theoretical expectations, a few anomalies merit further analysis:

Spectral Breakdown on web-NotreDame. Community collapse yielded $\mathrm{SpErr} = 3.20$ on web-NotreDame, far above its usual ≤ 0.96. Although collapse generally preserves quadratic forms for vectors constant on each community, here the summary size was only 570 nodes (CR ≈ 0.0016), so the neglected inter-community edges contribute an error term $\sum_{i,j \in C}(x_i - x_j)^2 w_{ij}$ that overwhelms the approximation. This suggests a practical lower bound on CR: when $|V_S|$ falls below the number of significant eigenmodes k, the bound $\mathrm{SpErr} = O\big(\sum_{j>k} \lambda_j\big)$ becomes vacuous, and fine-scale spectral structure cannot be recovered.

High Community Fidelity for Sparsifier on web-NotreDame. Contrary to its moderate NMI on larger graphs (≈ 0.73–0.86), the sparsifier achieved $\mathrm{NMI} = 0.844$ on the smallest dataset. Since effective-resistance sampling probability $p_e \propto w_e R_e$ favors intra-community edges, this suggests that web-NotreDame has tightly clustered modules with low average resistance. A low average degree within communities implies $R_{eff}(u, v) = O(1/d_{\text{avg}})$ is small, so sparsification almost entirely preserves community links – an effect less pronounced in sparser, larger graphs.

Coarsener Runtime Variability. Although spectral coarsening scales as $O(mk + nkT)$, its summarization time varied widely: 241 s on web-NotreDame versus 5 115s on web-BerkStan. This super-linear growth reflects both larger m and higher effective k (number of retained clusters). In practice, solver's runtime is sensitive to the spectral gap $\lambda_{k+1} - \lambda_k$; smaller gaps require more iterations to converge. The relatively small gap in web-BerkStan's spectrum thus explains its unusually long eigenvector computation.

Precision@50 Fluctuations for Collapse. Collapse's PageRank precision varied from 0.40 (web-NotreDame) to 0.84 (web-Stanford), more than expected given its uniform strategy of merging communities. This suggests that the distribution of PageRank mass across community sizes matters: when a hub's community is large, its relative rank within the supernode can drop significantly. Formally, if hub h with PageRank $\pi(h)$ is merged into a community C, its contribution becomes

$$\pi_S(C) = \sum_{v \in C} \pi(v),$$

so the normalized position of h in the top-k list depends on $\frac{\pi(h)}{\sum_{v \in C} \pi(v)}$. In graphs where hubs are embedded in communities of roughly uniform PageRank, collapse preserves top-k much better than when a single hub dominates a large module.

These unexpected findings highlight how real-world structural nuances – community tightness, spectral gaps, and PageRank distributions – can shift summarization behavior away from idealized bounds. They also suggest avenues for adaptive parameter tuning (e.g. controlling CR per community or adjusting k based on spectral gap estimates) to mitigate worst-case distortions.

6 Limitations

While our evaluation framework and implementations demonstrate strengths, several limitations should be acknowledged.

6.1 Approximation in MiniBatchKMeans

Our spectral-coarsening pipeline relies on `MiniBatchKMeans` to cluster nodes in the reduced eigenvector space. This choice greatly accelerates convergence on graphs with hundreds of thousands of nodes, but at the cost of clustering precision. By processing only small, randomly sampled batches and using a relatively lax convergence threshold, `MiniBatchKMeans` may converge to centroids that differ from the true global optimum. In practice, this can introduce noise into the spectral embedding partition, manifesting as increased inter-cluster edges, higher spectral error, and lower community alignment (NMI) than would be obtained with full-batch clustering. Although we observed stable results across multiple runs, practitioners seeking maximal fidelity should consider refining these clusters – either by initializing with a full-batch `KMeans` run, increasing batch sizes and iterations, or employing GPU-accelerated clustering libraries to balance speed with accuracy.

6.2 Loss of Edge Directionality

To leverage undirected spectral and community-detection techniques, all input graphs are symmetrized before summarization. While this enables the use of well-studied methods and simplifies Laplacian computation, it inevitably discards the inherent directionality of web-graph edges. As a result, summaries may misrepresent asymmetric relationships – such as Page A linking to Page B without reciprocity – and metrics that depend on edge orientation (e.g. directed PageRank, reachability analyses) may be affected. Although our undirected summaries still capture overall connectivity and community structure, future work should explore directed summarization approaches – such as directed Laplacian embeddings or flow-based community detection – to preserve navigational semantics without sacrificing the robustness of our framework.

6.3 Marginal Benefit of Sparsification on Sparse Web Graphs

Spectral sparsification is most beneficial when the original graph exhibits dense or redundant connectivity. However, web graphs typically maintain a sparse, nearly linear relationship between nodes and edges, as each page links to only a modest number of others. In this regime, further edge sampling yields diminishing compression gains – often at the expense of removing structurally important links. Consequently, the spectral sparsifier's CompR improvements on our web datasets were moderate, and aggressive sparsification risked fragmenting the graph. This suggests that in inherently sparse domains, node-aggregation or spectral-coarsening methods may offer more dramatic size reductions. Nevertheless, sparsification retains value as a lightweight, off-the-shelf summarizer that balances spectral and community fidelity for moderately sparse networks.

7 Conclusion and Future Work

In this work, we have presented a comprehensive framework for evaluating graph summarization methods along five key fidelity dimensions – spectral, community, distance, centrality, and compression – while also accounting for computational cost. By implementing and empirically comparing three representative paradigms (spectral sparsification, community-based collapse, and spectral coarsening) on four large-scale web graphs, we have both validated theoretical guarantees and revealed practical performance regimes. Our results demonstrate that:

- *Collapse* is the best in preserving spectral and distance properties under extreme size reduction, provided compaction is moderated to avoid loss of fine structure.
- *Sparsifier* offers the fastest end-to-end summarization with a balanced trade-off among spectral fidelity, community retention, and centrality preservation.
- *Coarsener* achieves the highest accuracy in identifying influential nodes and maintaining diffusion-based metrics, at the expense of higher runtime and increased spectral error.

These findings furnish practitioners with actionable guidelines for method selection based on target metrics and resource constraints, thus fulfilling our goal of a unified, application-driven summarization framework.

Building on this foundation, we envisage several directions to broaden and deepen the applicability of our framework:

Diverse Graph Domains. Extending evaluations to a wider spectrum of real-world networks – including social interaction graphs, citation networks, protein-interaction maps, and infrastructure grids – will test the robustness of method trade-offs under varying topologies, degree distributions, and community structures. Such studies could reveal domain-specific summarization strategies and inform parameter tuning for new applications.

Directed and Weighted Summaries. Incorporating edge directionality and weight heterogeneity remains a key challenge. Future work could adapt directed Laplacian constructions or flow-based community detection to generate summaries that preserve both structural and navigational semantics. Similarly, handling multi-edge or temporal weights would enable faithful summarization of dynamic or multiplex networks.

Hybrid and Multi-Stage Pipelines. Our experiments indicate that no single paradigm universally dominates all fidelity metrics. Designing hybrid pipelines – where, for example, sparsification is followed by community collapse, or coarsening is applied only within detected modules – may yield summaries that harness complementary strengths. Research into the optimal composition and sequencing of summarization stages could unlock new performance frontiers.

Scalable, High-Precision Clustering. Improving the accuracy of spectral coarsening without prohibitive cost is critical. Integrating GPU-accelerated or distributed clustering libraries (e.g. cuML, Spark MLlib) and exploring advanced initialization or refinement schemes (e.g. spectral-seeded KMeans) can reduce approximation error in large-scale eigenvector clustering.

Streaming and Dynamic Graphs. Many real-world networks evolve over time. Extending our framework to support incremental summarization – updating summaries in response to edge or node changes with sublinear cost – would enable real-time analytics on social media, communication logs, and IoT streams. Techniques from dynamic spectral approximation and online community detection could be leveraged here.

Theoretical Analysis of Hybrid Methods. While our current theoretical expectations focus on standalone paradigms, formalizing error and runtime bounds for hybrid summarizers represents an open problem. Developing unified approximation guarantees for combined sampling-aggregation or clustering-refinement workflows will provide deeper insights into the fundamental limits of graph summarization.

By pursuing these directions, we aim to evolve our unified evaluation framework into a comprehensive toolset that guides both theoretical research and practical deployments of graph summarization across diverse and dynamic networked systems.

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